

Potential Energy Surface and State-to-State Dynamics of Chemical Reactions

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- 4 **Chapter 4:** Construction of the global potential energy surface for the ground electronic state of HeLiH^+ system
- 5 **Chapter 5:** State-to-state dynamics of isotopic exchange reaction $^{18}\text{O} + ^{16}\text{O}^{16}\text{O} (v = 0, j = 1) \rightarrow ^{18}\text{O}^{16}\text{O} + ^{16}\text{O}$
- 6 **Chapter 6:** Summary and Outlook

1 Global fitting of potential energy surface (PES)

- Machine learning (ML) methods
- Non-ML methods

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- Machine learning (ML) methods
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2 Adiabatic reaction dynamics studies

- 1 Quantum mechanical method
- 2 Classical method

Introduction

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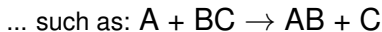


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Molecular reaction dynamics deals with the molecular level understanding of a reaction.

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Approaches: Theoretical and Experimental.

Theoretical approach: Theory and Methodology

In atomic dimensions,

Time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi(\vec{r}_e, \vec{r}_N, t)}{\partial t} = \hat{H} \psi(\vec{r}_e, \vec{r}_N, t). \quad (1)$$

Time-independent Schrödinger equation:

$$\hat{H} \psi(\vec{r}_e, \vec{r}_N) = E \psi(\vec{r}_e, \vec{r}_N) \quad (2)$$

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*Solving the above two equations is not easy, unless we invoke
Born-Oppenheimer (BO) approximation!*

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Adiabatic dynamics is used in the works presented here!

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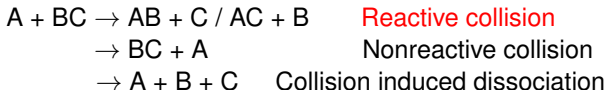
Since electronic structure calculations are expensive, fitting and using the analytical form is the best solution!

Theory and Methodology: Introduction to dynamics

Reaction system in current work: Triatomic reaction system (atom-diatom collision)

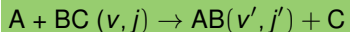
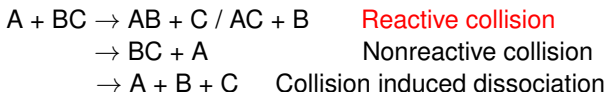
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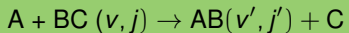
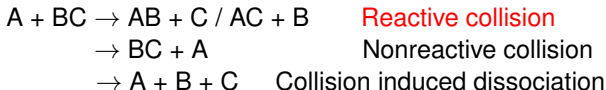
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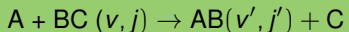
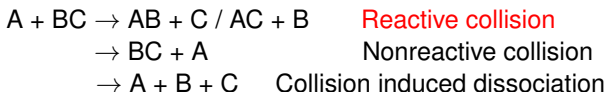
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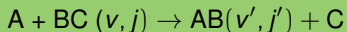
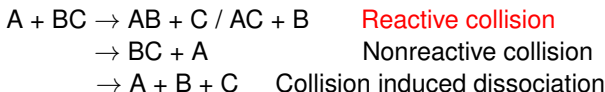
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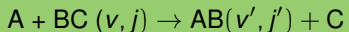
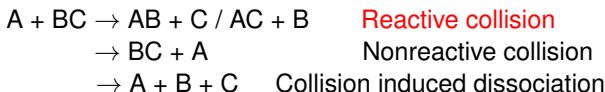
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QM and Classical both methods are used in our investigations!

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Eq. 9 is the complex but the one used in the current works is real form.

Equations of motion (Hamilton's equation):

$$\frac{\partial q}{\partial t} = \frac{\partial H}{\partial p} = \frac{p}{\mu} \quad (10)$$

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Form of classical method used:

Quasi classical trajectory (QCT) method: Classical method with quantum approximations.

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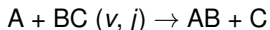
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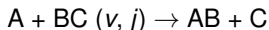


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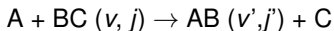
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- ▶ PES fitting has two parts:
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The two parts are equally important!

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- ▶ **Generating energy points involve:**
 - Choice of electronic structure methods.
 - Choice of coordinate grids.

Theory and Methodology: Fitting with analytical function

- ▶ Representing the potential
 - Many-body expansion (MBE)¹
 - Double many-body expansion
 - Global representation

¹ *Molecular Physics*, **29**(5):1387–1407, 1975.

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- **Many-body expansion (MBE):**

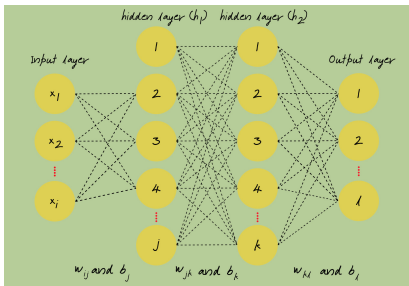
$$V_{ABC}^{tot,MBE}(r_{AB}, r_{BC}, r_{AC}) = \sum_i V_i^{(1)} + \sum_j V_j^{(2)}(r_j) + V^{(3)}(r_{AB}, r_{BC}, r_{AC}) \quad (13)$$

$$\sum_i V_i^{(1)} = V_A^{(1)} + V_B^{(1)} + V_C^{(1)}$$

$$\sum_j V_j^{(2)}(r_j) = V_{AB}^{(2)}(r_{AB}) + V_{BC}^{(2)}(r_{BC}) + V_{AC}^{(2)}(r_{AC})$$

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Theory and Methodology: ML method



► Neural network (NN)

- Hyperparameters: No. of neurons and layers, learning rate, etc., affect the training.
- Overfitting and underfitting are the primary bottlenecks.

NN function (with one neuron in the output layer):

$$V^{\text{NN}} = b^{(3)} + \sum_{k=1}^{N_k} w_k^{(3)} f^{(2)} \left[b_k^{(2)} + \sum_{j=1}^{N_j} w_{jk}^{(2)} f^{(1)} \left(b_j^{(1)} + w_{ij}^{(1)} G_i \right) \right] \quad (14)$$

The ML method cares about the numbers without knowing the origin of the behavior of them!

Data should **best** represent the global configuration space of the molecular system.

Theory and Methodology: Non ML method

► Aguado form:¹

¹*J. Chem. Phys.* **96**, 1265–1275, 1992.

Theory and Methodology: Non ML method

► Aguado form:¹

Two-body part:

$$V_{AB}^{(2)}(R_{AB}) = \frac{c_0 e^{-\alpha_{AB} R_{AB}}}{R_{AB}} + \sum_{i=1}^N c_i \rho_{AB}^i \quad (15)$$

$$\rho_{AB} = R_{AB} e^{-\beta_{AB}^{(2)} R_{AB}}$$

Three-body part:

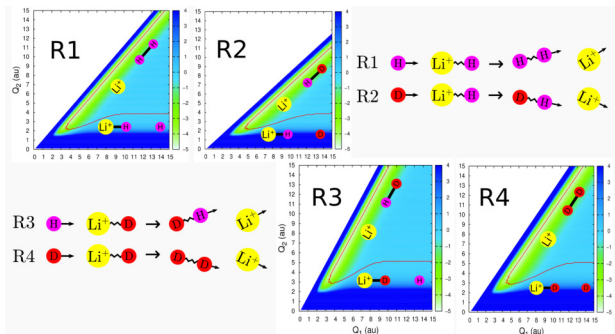
$$V_{ABC}^{(3)} = \sum_{i,j,k=1}^M d_{ijk} \rho_{AB}^i \rho_{BC}^j \rho_{AC}^k \quad (16)$$

$$i + j + k \neq i \neq j \neq k \text{ and } i + j + k \leq M$$

¹J. Chem. Phys. **96**, 1265–1275, 1992.

Chapter 3

Quantum and classical comparison and Isotopic effect in $\text{H} + \text{LiH}^+ \rightarrow \text{H}_2 + \text{Li}^+$ reaction^{1 2 3}



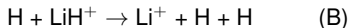
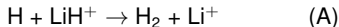
¹J. Phys. Chem. A **127**, 51, 10733 (2023).

²J. Phys. Chem. A **125**, 16, 3387 (2021).

³Phys. Chem. Chem. Phys. **23**, 27327 (2021).

Introduction and Motivation

- ▶ **Early universe:** Li, H, He, and D.
- ▶ **LiHe⁺ and LiH⁺:** possible molecular forms of Li in early universe.
- ▶ Destruction and formation of these molecular species: important reaction events.
- ▶ **Motivation:** possible collision of LiH⁺ with abundant H atoms.
- ▶ **Examples:**



A: Exchange reaction. **B:** Collision-induced dissociation.

The reaction considered in the present work:



Research themes

- 1 Comparison of quantum and classical methods
- 2 Study of the isotopic effect

¹J. Chem. Phys. **119**, 11241 (2003).

²J. Phys. Chem. A **125**, 16, 3387 (2021).

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PES: Martinazzo et al. ¹

Theoretical methodology: QCT-HB/GB methods

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Research themes

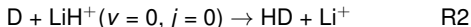
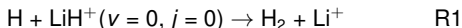
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Theoretical methodology: QCT-HB/GB methods

Comparison of quantum and classical methods

Reactions considered:

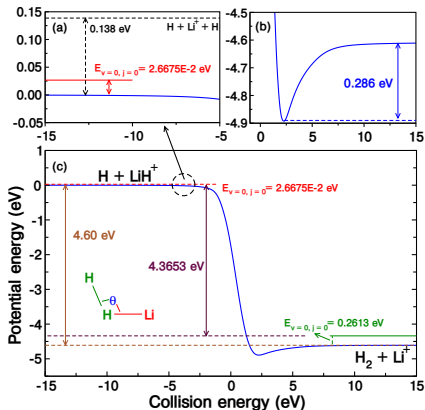
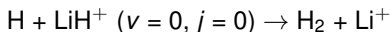


Note: QM results from earlier² & QCT results from current work.

¹J. Chem. Phys. **119**, 11241 (2003).

²J. Phys. Chem. A **125**, 16, 3387 (2021).

H + LiH⁺ → H₂ + Li⁺ reaction: A brief introduction



- **Exoergicity:** ~ 4.60 eV (without ZPE) and ~ 4.36 eV (with ZPE of reactant and product diatoms)
- CID opens at ~ 0.138 eV.
- **Shallow minima:** ~ 0.286 eV below product asymptote.

QM and QCT comparison: Total reaction probability

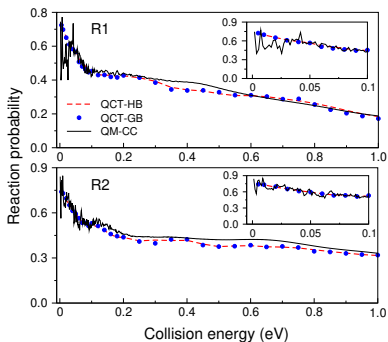
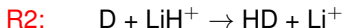


Figure 1: Total reaction probability at $J=0$.

QM and QCT comparison: Total reaction probability

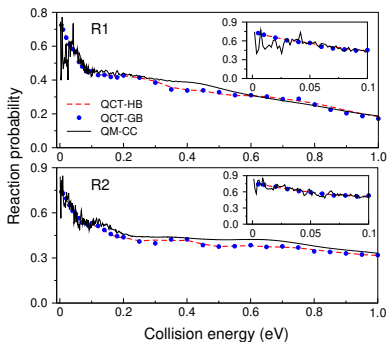
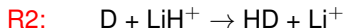


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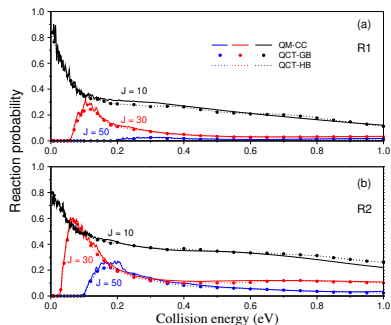


Figure 2: Total reaction probability at different J values.

QM and QCT comparison: ICS

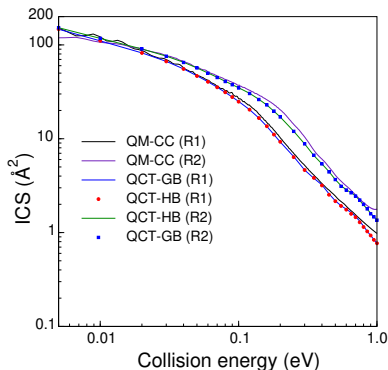
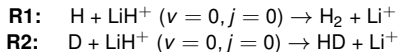


Figure 3: Initial state-selected QM and QCT ICSs for R1 and R2.



- ICS decreases with increasing collision energy: *typical barrierless exoergic reaction behavior.*
- QCT-GB/HB and QM ICSs are following each other.
- QM shows resonance structure which are not available in QCT ICS.

QM and QCT comparison: DCS

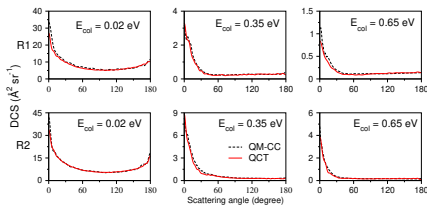


Figure 4: QCT DCSs for R1 and R2.

- DCS is forward bias, suggesting a direct mechanism.
- At low collision energy, backward scattering is observed.

QM and QCT comparison: DCS

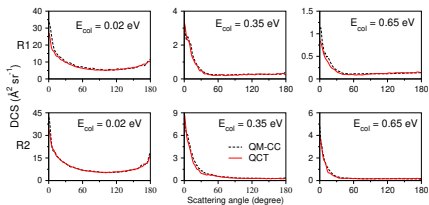


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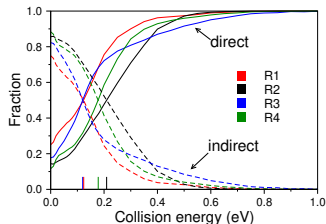


Figure 5: Fraction of indirect and direct trajectories for R1 and R2.

QM and QCT comparison: Rate constants

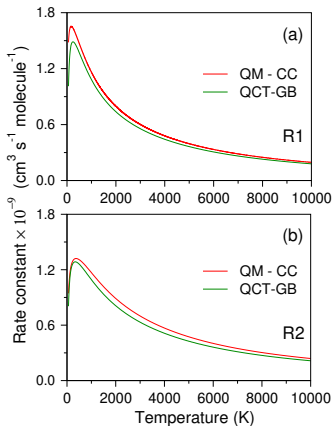
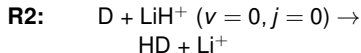
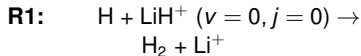


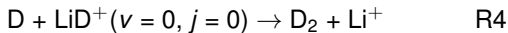
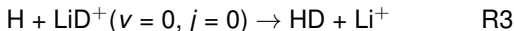
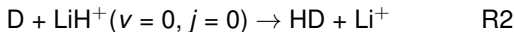
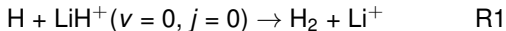
Figure 6: State-selected rate constants for ($v=0, j=0$) initial state of reactant diatom.



- Rate constants are shown up to 10000 K.
- Rate constant increases first and decreases later.
- The QM and QCT rate constants agree with each other.

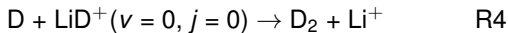
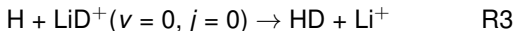
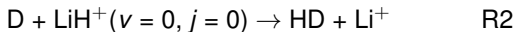
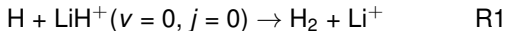
Part 2 : Isotopic effect

Reactions considered in the study of isotopic effect:



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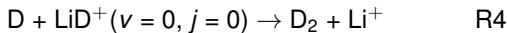
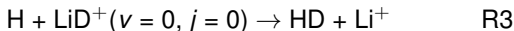
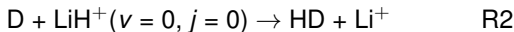
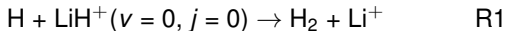
Reactions considered in the study of isotopic effect:



Only QCT methods are used for isotopic effect investigation. Why?

Part 2 : Isotopic effect

Reactions considered in the study of isotopic effect:



Only QCT methods are used for isotopic effect investigation. Why?

Good agreement between QCT and QM results for R1 and R2, increases the confidence in the accuracy of QCT method.

Isotopic effect ?

Isotopic effect: *change in the reaction dynamics or observables by replacing the isotopic variant in a reaction.*

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How do isotopes affect the dynamics of a reaction?

- ▶ Changing the isotopes is changing the masses of the nuclei taking part in the reaction.
- ▶ The adiabatic electronic potentials for the isotopic variants remain the same. However, the forces change as the masses are different.

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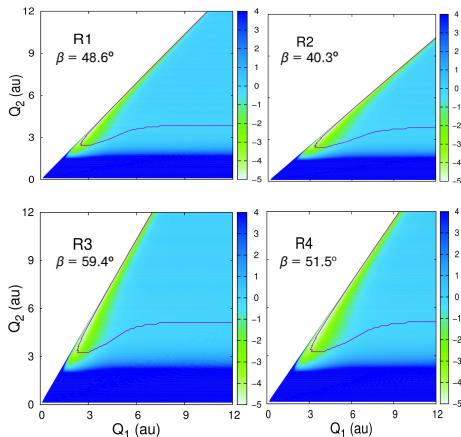
How do isotopes affect the dynamics of a reaction?

- ▶ Changing the isotopes is changing the masses of the nuclei taking part in the reaction.
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The PESs remain the same for different isotopic variants but are perceived differently!

Is there a way to observe the effect of isotopes on the potential surface?
Possibly, yes! Mass-weighted PES.

Isotopic effect: Mass-weighted PES



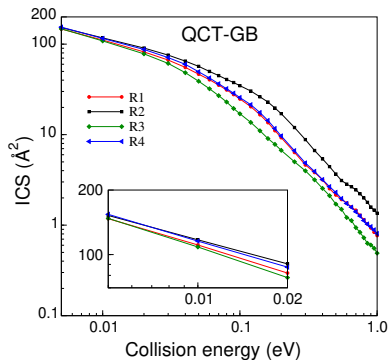
The reactant and product asymptote are skewed for different isotopic variants.

Skewness: $R2 > R1 \sim R4 > R3$

Skewness is correlated with the reaction observables!!

Figure 7: Contour map of the mass-weighted PESs and MEPs (shown as a violet-colored line) at an attacking angle ($\angle H_A H_B Li$).

Isotopic effect on ICS



R1: $\text{H} + \text{LiH}^+$ **R2:** $\text{D} + \text{LiH}^+$

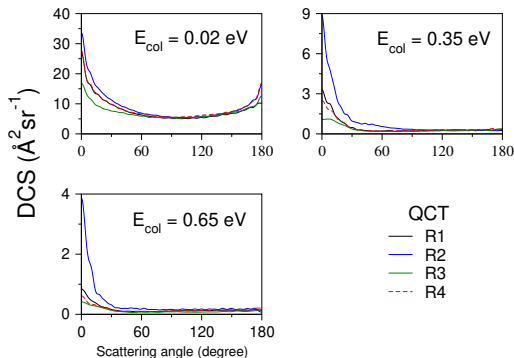
R3: $\text{H} + \text{LiD}^+$ **R4:** $\text{D} + \text{LiD}^+$

Order of ICS: $\text{R2} > \text{R1} \sim \text{R4} > \text{R3}$

Skewness: $\text{R2} > \text{R1} \sim \text{R4} > \text{R3}$

Figure 8: Initial state-selected QCT ICSs for R1, R2, R3, and R4.

Isotopic effect on DCS



R1: $\text{H} + \text{LiH}^+$ **R2:** $\text{D} + \text{LiH}^+$

R3: $\text{H} + \text{LiD}^+$ **R4:** $\text{D} + \text{LiD}^+$

- Isotopic effect is quantitative.
- The quantitative order of DCS: correlated with the skewness.

Figure 9: Initial state-selected QCT DCSs for R1, R2, R3, and R4.

Chapter 3: Conclusion

QM and QCT comparison:

- ▶ Good agreement at state-selected and state-to-state level.
 - ▶ Scalar and vector observables both are in good agreement.
-

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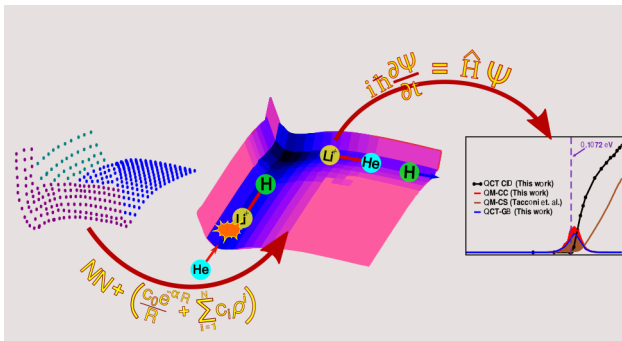
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Isotopic effect:

- ▶ The treatment of different isotopic variants **can be visualised in mass-weighted PES**.
- ▶ **Significant isotopic effect** observed.
- ▶ Isotopic effect: **correlated with the skewness** of the reactant-product channels.

Chapter 4

Construction of the global potential energy surface for the ground electronic state of HeLiH^+ system and dynamics of $\text{He} + \text{LiH}^+ \rightarrow \text{LiHe}^+ + \text{H}$ reaction¹



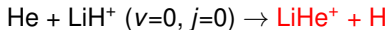
¹J. Chem. Phys. **161**, 124308 (2024).

Introduction and Motivation

- ▶ Li is one of the elements present in the early universe.
- ▶ LiH^+ is possible diatomic form of Li.
- ▶ **Motivation:** possibility of He and LiH^+ collision.
- ▶ Ground electronic state of LiHeH^+ facilitates the following reactions:
 - $\text{He} + \text{LiH}^+ \rightarrow \text{LiHe}^+ + \text{H}$ or $\text{Li}^+ + \text{He} + \text{H}$
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 - $\text{H} + \text{LiHe}^+ \rightarrow \text{LiH}^+ + \text{He}$ or $\text{Li}^+ + \text{He} + \text{H}$
- ▶ Reaction considered in current work:



Brief literature summary and way forward ...

- ▶ **Only one PES** (Wernli et al.¹) is available. Method: MRCI
- ▶ **Only report** (Tacconi et al.²) on the dynamics: $\text{He} + \text{LiH}^+ \rightarrow \text{LiHe}^+ + \text{H}$ and $\text{H} + \text{LiHe}^+ \rightarrow \text{LiH}^+ + \text{He}$
- ▶ **Results of Tacconi et al. used in assessment of LiHe^+** (Bovino et al.³) in early universe.

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Earlier dynamical results doesn't follow the usual trend,
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Current study?

- ▶ **New dynamics results.**
- ▶ **Constructing HeLiH^+ surface** using high level data point.

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Construction of PES: Representing the potential of HeLiH^+

Many body expansion (MBE)¹ is used to represent the PES.

$$V_{\text{LiHeH}^+}^{\text{tot,MBE}}(r_{\text{LiH}^+}, r_{\text{LiHe}^+}, r_{\text{HeH}}) = \sum_i v_i^{(1)} + \sum_j v_j^{(2)}(r_j) + V^{(3)}(r_{\text{LiH}^+}, r_{\text{LiHe}^+}, r_{\text{HeH}}) \quad (17)$$

$$\sum_i v_i^{(1)} = v_{\text{Li}^+}^{(1)} + v_{\text{He}}^{(1)} + v_{\text{H}}^{(1)} = 0$$

$$\sum_j v_j^{(2)}(r_j) = v_{\text{LiH}^+}^{(2)}(r_{\text{LiH}^+}) + v_{\text{LiHe}^+}^{(2)}(r_{\text{LiHe}^+}) + v_{\text{HeH}}^{(2)}(r_{\text{HeH}})$$

¹ *Molecular Physics*, **29**(5):1387–1407, 1975.

Construction of PES: Generating ab initio data and fitting methods

Generating ab initio data:

- ▶ **Grid:** Two bond distances (r_{HeLi^+} and r_{LiH^+}) and bond angle between these (θ).
 - ▶ **Electronic structure calculation:** Full configuration interaction (FCI) method + aug-ccpVQZ basis + extrapolation to CBS limit.
-

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Fitting method:

- ▶ **Two-body part:** Neural Network (NN) algorithm + Aguado Paniagua (AP) form. **AP:** HeH, **NN:** LiHe⁺ and LiH⁺
- ▶ **Three-body part:** AP
- ▶ **Optimization algorithm:** Levenberg-Marquardt (LM)

Fitting errors for two-body and three-body parts

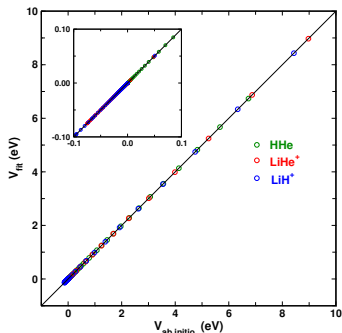


Figure 10: Fitted vs ab initio energy plot for diatoms.

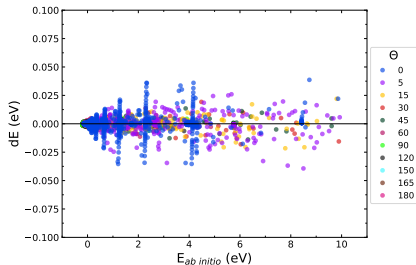


Figure 11: Fitting error ($E_{fit} - E_{ab\ initio}$) at each ab initio data point for overall fitting of LiHeH^+ system.

Root mean square error (RMSE) of fitting:

Two-body (LiHe^+ , LiH^+ , and HeH)
Three-body
Overall

1.04E-04 eV, 3.4E-05 eV, and 9.8E-06 eV
 1.63E-03 eV (data < 10.0 eV)
 9.28E-04 eV (data < 2.0 eV)

Potential energy curves

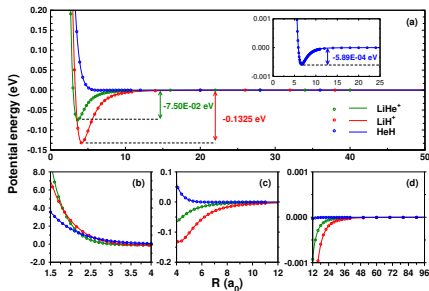


Figure 12: Fitted vs ab initio potential energy curves for diatoms.

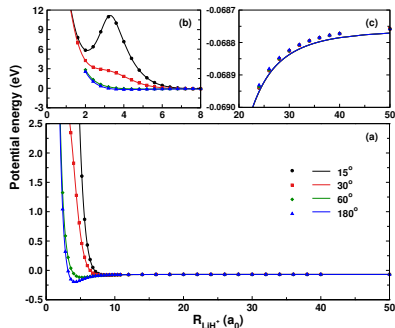


Figure 13: Fitted vs ab initio potential cuts when LiHe^+ is fixed at $3.40 a_0$.

Fitted and ab initio potential energy surfaces

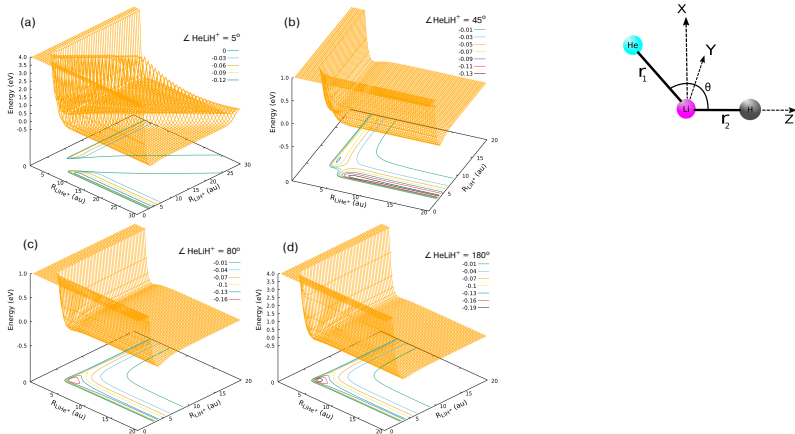


Figure 14: PESs at different attacking angles ($\angle \text{HeLiH}^+$).

Fitted and ab initio potential energy surfaces

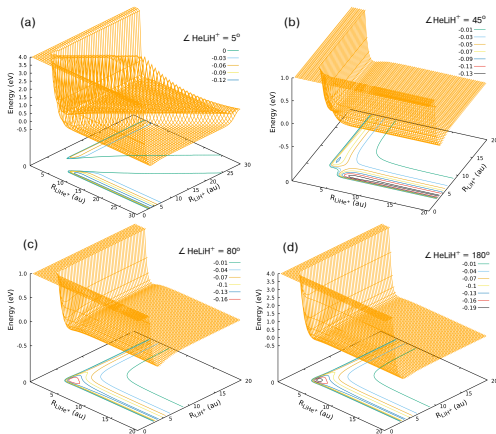


Figure 14: PESs at different attacking angles ($\angle \text{HeLiH}^+$).

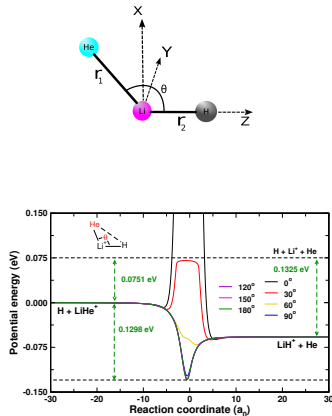
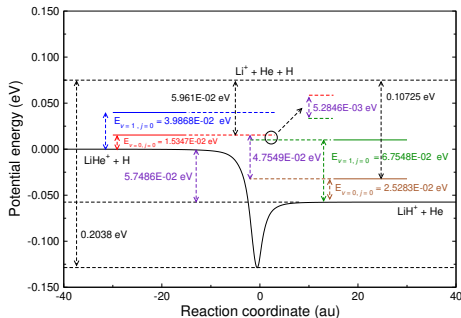


Figure 15: MEPs at different attacking angles ($\angle \text{HeLiH}^+$).

Energetics



- **R1:** $\text{H} + \text{LiHe}^+ \rightarrow \text{LiH}^+ + \text{He}$
- **R2:** $\text{He} + \text{LiH}^+ \rightarrow \text{LiHe}^+ + \text{H}$
- **Collision induced dissociation (CID):**
 $\text{He} + \text{LiH}^+ \rightarrow \text{Li}^+ + \text{He} + \text{H}$

A little more about (R2)= $\text{He} + \text{LiH}^+ \rightarrow \text{LiHe}^+ + \text{H} \dots$

- $\Delta E = 4.7549\text{E-}02 \text{ eV}$ (including ZPE). **Endoergic.**
- CID opens at 0.10725 eV.

For reaction R2, CID opens at a very low collision energy of ~ 0.1075 eV, affecting the dynamics of the reaction!

Theoretical methods:

1 TDQM

involves the convergence of input parameters, calculations of S-matrix elements for different total angular momentum (J) collisions, and calculations of reaction observables.

2 QCT

involves the convergence of b_{max} values, running trajectories, and calculations of reaction observables.

Results: Reaction probabilities

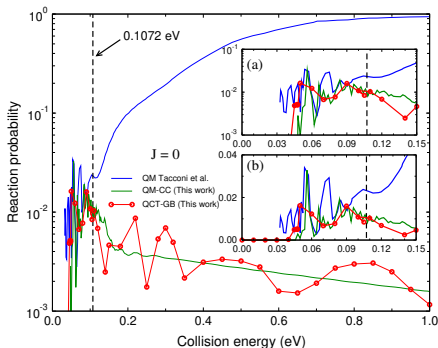


Figure 16: Reaction probability at $J = 0$ for $\text{He} + \text{LiH}^+$ ($v=0, j=0$) $\rightarrow \text{LiHe}^+ + \text{H}$ reaction.

- QM and QCT-GB results increase till ~ 0.1072 eV and then decrease with collision energy.
- Results from Tacconi et al.¹ keep increasing even after the opening of CID.

¹Phys. Chem. Chem. Phys. 14, 673 (2012).

Integral cross section (ICS)

How does the CID affect the reactivity of a reaction?

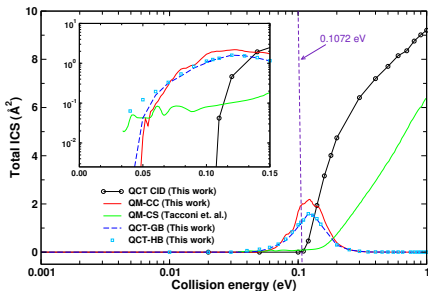


Figure 17: ICSs for $\text{He} + \text{LiH}^+(v=0, j=0) \rightarrow \text{LiHe}^+ + \text{H}$ reaction

- QM-CC and QCT ICSs from our work are compared with QM-CS ICS of Tacconi et al.¹
- QCT ICS for CID channel from our work is also shown.
- **Effect of the CID channel is evident!**

¹Phys. Chem. Chem. Phys. 14, 673 (2012).

State-specific rate constants

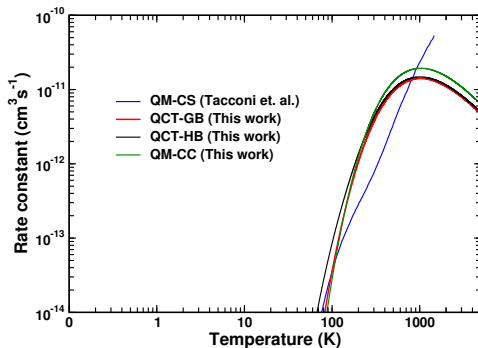


Figure 18: State-specific rate constant for $\text{He} + \text{LiH}^+ (\nu=0, j=0) \rightarrow \text{LiHe}^+ + \text{H}$ reaction.

- QM-CC and QCT, both rate constants, are calculated in our work.
- The state-specific rate constants from our work are compared with the results of Tacconi et al.¹

¹Phys. Chem. Chem. Phys. 14, 637 (2012).

Fitting of PES:

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Fitting of PES:

- ▶ A **new global PES** was constructed.
 - ▶ Data points: generated at the **FCI level of theory**, much higher than the earlier surface.
 - ▶ Overall **RMSE: 14.21 cm^{-1}** (data $\leq 10 \text{ eV}$) and **7.48 cm^{-1}** (data $\leq 2 \text{ eV}$).
 - ▶ Neural network and Aguado fitting forms were used in combination.
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Dynamics of the reaction:

- ▶ Dynamics of **$\text{He} + \text{LiH}^+ \rightarrow \text{LiHe}^+ + \text{H}$** is investigated using QM and QCT method.
- ▶ The reaction observables show the **effect of the opening of the CID** channel at 0.1075 eV .
- ▶ Our **results show a different trend** than the results of Tacconi et al.¹ reported earlier.

¹Phys. Chem. Chem. Phys. 14, 673 (2012).

Chapter 5

State-to-state dynamics of isotopic exchange reaction $^{18}\text{O} +$
 $^{16}\text{O}^{16}\text{O} (v = 0, j = 1) \rightarrow ^{18}\text{O}^{16}\text{O} + ^{16}\text{O}^1$

¹manuscript under preparation.

Introduction

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Historically, major interest in this reaction comes from the importance of O_3

Introduction

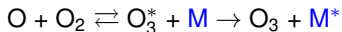
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- ▶ **Mass-independent fractionation** leading to enrichment of heavier isotope was observed in O_3 (1980-83).

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- ▶ One of the mechanisms of O₃ formation is **Lindemann mechanism**:

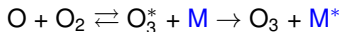


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- ▶ **Mass-independent fractionation** leading to enrichment of heavier isotope was observed in O₃ (1980-83).

- ▶ One of the mechanisms of O₃ formation is **Lindemann mechanism**:



- ▶ Isotopic exchange reaction ($^a\text{O} + ^b\text{O}^c\text{O} \rightarrow \text{O}_3^* \rightarrow ^b\text{O} + ^a\text{O}^c\text{O}$) competes with the O₃ formation.
- ▶ **Combination of different isotopes**: key to understand MIF.

Brief literature summary: Where do we stand?

- Many PESs have been constructed over the years! (2001-2017 and so on)

¹J. Chem. Phys. 142, 17, 174311 (**2015**).

²J. Chem. Phys. 142, 17, 174312 (**2015**).

³J. Phys. Chem. Lett. 9, 8, 1931 (**2018**).

⁴J. Chem. Phys. 139, 134307, (**2013**).

⁵Chem. Phys. Lett. 776, 138648, (**2021**).

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- **2017:** *First exact match of theoretical and experimental data.* Guillon et al.³ (Reaction: 8+66. PES: Tyuterev et al.⁴)

Reason for the mismatch in earlier investigations: An error in the form of a submerged barrier.

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Reason for the mismatch in earlier investigations: An error in the form of a submerged barrier.

- **2021:** Korutla et al.⁵: State-to-state DCS and ICSs reported.

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²J. Chem. Phys. 142, 17, 174312 (2015).

³J. Phys. Chem. Lett. 9, 8, 1931 (2018).

⁴J. Chem. Phys. 139, 134307, (2013).

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What else is left?

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Theoretical methodology:

- ▶ TDWP method in energy range: 0.0 eV - 0.1 eV.
 - ▶ Only 8+66 isotopic variant of the reaction is investigated.
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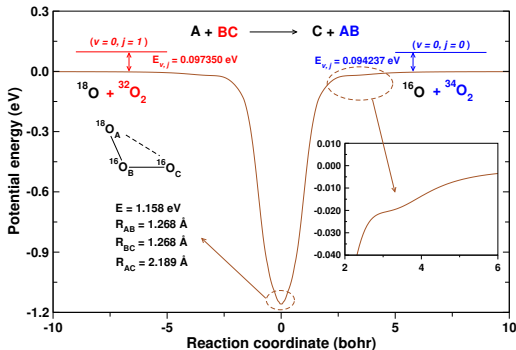
- ▶ TDWP method in energy range: 0.0 eV - 0.1 eV.
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-

PES:

- ▶ Varga et al.¹ (*surface doesn't have the false submerged barrier.*)

¹ *J. Chem. Phys.*, 147, 154312 (2017).

Energetics



- **8+66 reaction:**
Exoergic $\sim 0.003113 \text{ eV}$ (including ZPE of product and ($v=0, j=1$) of reactant).
- **Deep minima** of energy 1.158 eV is present.
- **No submerged reef** (check in-set figure).

Results: Total reaction probability

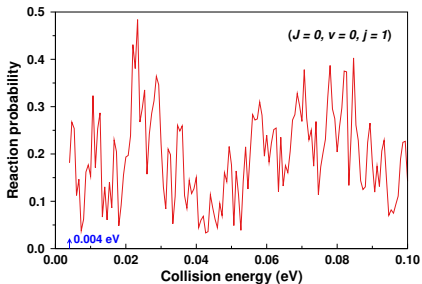


Figure 19: Total reaction probability at $J=0$.

Results: Total reaction probability

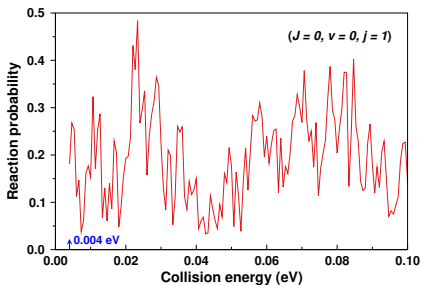


Figure 19: Total reaction probability at $J=0$.

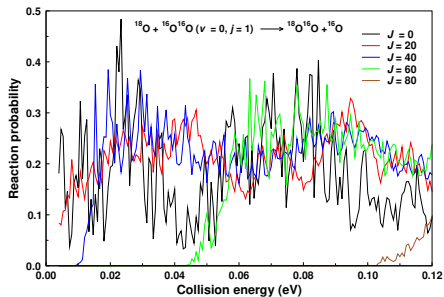


Figure 20: Total reaction probability at selected J .

Results: State-selected integral cross section

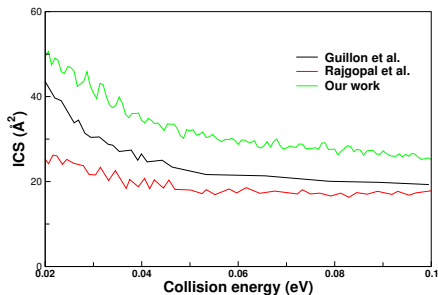
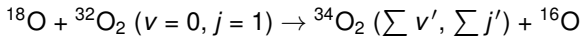


Figure 21: ICSs as a function of collision energy.

- ▶ ICS ↓ as collision energy ↑.
- ▶ The three ICSs follow the same pattern.

Results: State-to-state ICS

Total energy: collision energy + $E_{v,j}$ of $^{32}\text{O}_2$ (0.09735 eV).

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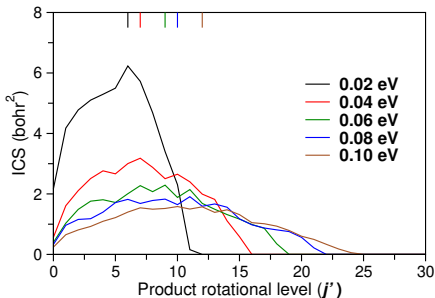


Figure 22: ICSs vs product rotational level (j'') at different collision energies.

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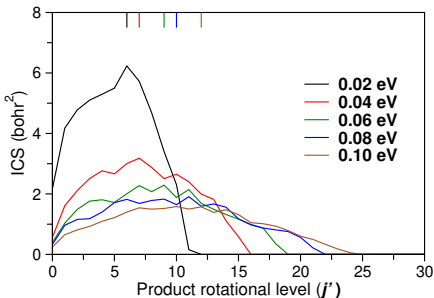


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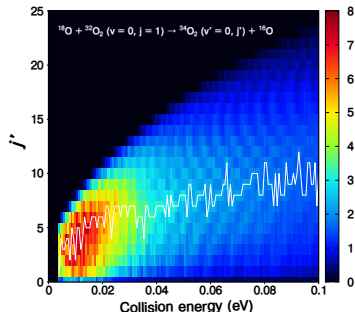


Figure 23: ICSs vs product rotational level (j') as a function of collision energies.

- Collision energy ($\uparrow$): ICS distributions become rotationally hot.
- Collision energy ($\uparrow$): broader range of j' .

Conclusion

General findings:

- ▶ Rich resonances are found in reaction probability and ICSs for the complete collision energy range of investigation.
 - ▶ ICSs follow the same behavior as the results reported earlier.
-

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- ▶ Current work presents detailed TDWP investigation of 8+66 reaction variant.
 - ▶ Detailed dynamical insights about the reaction.
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Scope of current work:

- ▶ Current work presents detailed TDWP investigation of 8+66 reaction variant.
- ▶ Detailed dynamical insights about the reaction.

Limitations:

- ▶ Computation time is the bottleneck: **Current QM calculations took ~ 23 years of computational time.**
- ▶ So, it is difficult to comment on thermal rate constants and isotopic effects using current calculations.

Chapter 6: Summary and Outlook

- ▶ **Two different research themes:** construction of PES and dynamics.
 - ▶ **Chapter 3 and 4:** Two pathways of LiH^+ destruction are investigated.
 - ▶ QCT methods are accurate enough for exoergic barrierless reaction system.
 - ▶ **Chapter 5:** Rigorous and accurate dynamics is performed for 8+66 variant of $\text{O} + \text{O}_2$ reaction.
-

¹manuscript under preparation.

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What's next?

¹manuscript under preparation.

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What's next?

- ▶ To construct the PESs for the molecular system with atoms >3 .
- ▶ A few problems were encountered while using ML, which has to be perfected. Since ML methods are extremely helpful in constructing PES with large no. of atoms.
- ▶ Report the formation of LiH^+ on HeLiH^+ channel ¹.
- ▶ To investigate the origin of resonances observed in 8+66 reaction observables.

¹manuscript under preparation.

Acknowledgement



Supervisor:

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UPE II, School of Chemistry, UoH

Funding:

UoH and DST

Lab mates, friends, and family

Visit:  <https://github.com/ajaymrawat/>

Thank you!



Additional slides

Chapter 2: Extras

Wavepacket propagation in Chebyshev approximation:

$$q^{J,\Omega'}(r_N, t + \Delta t) = 2\hat{H}_s q^{J,\Omega'}(r_N, t) - q^{J,\Omega'}(r_N, t - \Delta t) \quad (18)$$

$$q^{J,\Omega'}(t) = \hat{H}_s q^{J,\Omega'}(t - \Delta t) - \sqrt{1 - \hat{H}_s^2} p^{J,\Omega'}(t - \Delta t) \quad (19)$$

$\hat{H}_s$ is scaled Hamiltonian and can be written as $\hat{H}$ easily.

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Hamiltonian for A+BC reaction system:

$$\begin{aligned} \hat{H} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) = & -\frac{\hbar^2}{2\mu_{r_c}} \frac{\partial^2}{\partial r_c^2} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) - \frac{\hbar^2}{2\mu_{R_c}} \frac{\partial^2}{\partial R_c^2} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \\ & + \left(\frac{1}{2I_{r_c}} + \frac{1}{2I_{R_c}} \right) \hat{J}^2 q^{J,\Omega'}(R_c, r_c, \gamma_c, t) + \frac{(\hat{J}^2 - 2\hat{J}_z \hat{J}_z)}{2I_{R_c}} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \\ & + \hat{V}_{elec} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) - \frac{(\hat{J}_+ \hat{J}_- + \hat{J}_- \hat{J}_+)}{2I_{R_c}} q^{J,\Omega'}(R_c, r_c, \gamma_c, t) \quad (20) \end{aligned}$$

Action of different parts of Hamiltonian: carried out with different numerical methods.

► **Reaction probability:**

$$P_{vj}^J(E) = \sum_{\Omega=0}^j \frac{g_{\Omega}}{2j+1} \sum_{v'} \sum_{j'} \sum_{\Omega'} \frac{1 + \delta_{\Omega'0}}{2} \sum_{p=0}^1 |S_{v,j\Omega \rightarrow v'j'\Omega'}^{J,p}(E)|^2 \quad (21)$$

$$P_{vj}^J(E) = \sum_{\Omega} \frac{g_{\Omega}}{2j+1} \sum_{v'} \sum_{j'} \sum_{\Omega'} |S_{v,j\Omega \rightarrow v'j'\Omega'}^J(E)|^2 \quad (22)$$

$g_{\Omega} = 1$ for $\Omega = 0$ and $g_{\Omega} = 2$ for $\Omega > 0$.

► **Cross section (ICS):**

$$\sigma_{vj}(E) = \frac{\pi}{k_{vj}^2} \sum_{\Omega=0}^j \frac{g_{\Omega}}{2j+1} \sum_{J \geq \Omega}^{J_{\max}} (2J+1) \sum_{v'} \sum_{j'} \sum_{\Omega'}^{\min(J,j')} P_{vj\Omega \rightarrow v'j'\Omega'}^J(E) \quad (23)$$

The observables are state-selected. State-to-state observables can also be calculated.

► Rate Constant:

$$k_{vj}(T) = \sqrt{\frac{8k_B T}{\pi \mu_{A-BC}}} \frac{1}{(k_B T)^2} \int_0^\infty \sigma_{vj}(E) e^{-E/k_B T} E dE \quad (24)$$

► Differential cross section (DCS):

$$\sigma_{vj \rightarrow v'j'}^{\text{DCS}}(E, \theta) = \frac{1}{4k_{vj}^2} \sum_{\Omega=0}^j \left(\frac{g_\Omega}{2j+1} \right) \sum_{J \geq \Omega}^{J_{\max}} (2J+1) \sum_{\Omega'} |S_{v,j\Omega \rightarrow v'j'\Omega'}^J(E) d_{\Omega\Omega'}^J(\pi - \theta)|^2 \quad (25)$$

Theory and Methodology: QCT observables

► Reaction probability:

$$P_{vj}^{\text{GB}}(E_{\text{col}}) = \frac{\sum_i^{N_r} w_i}{N} \quad P_{vj}^{\text{HB}}(E_{\text{col}}) = \frac{N_r}{N} \quad (26)$$

► Cross section (ICS):

$$\sigma_{vj \rightarrow v'j'}^{E_{\text{col}}} = \pi b_{\text{max}}^2 P_{vj \rightarrow v'j'}(E_{\text{col}}) \quad (27)$$

► Rate constant:

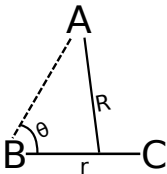
$$k_{vj \rightarrow v'j'}(T) = \sqrt{\frac{8k_B T}{\pi \mu_{\text{A-BC}}}} \left(\frac{1}{k_B T} \right)^2 \int_0^\infty E_{\text{col}} \sigma_{vj \rightarrow v'j'}^{E_{\text{col}}} e^{-\frac{E_{\text{col}}}{k_B T}} dE_{\text{col}} \quad (28)$$

► Differential cross section (DCS):

$$\frac{d\sigma_{vj}(E_{\text{col}}, \theta)}{d\omega} = \frac{\sigma_{vj}^{E_{\text{col}}}}{2\pi} \left[\frac{1}{2} + \sum_{n=1}^M a_n P_n(\cos\theta) \right] \quad (29)$$

$$a_n = \frac{2n+1}{2} \frac{1}{\sum_{i=1}^{N_r} w_i} \sum_i^{N_r} P_n(\cos\theta_i)$$

Isotopic effect: Mass-weighted PES



- Mass-weighted coordinates Q_1 and Q_2 are:

$$Q_1 = \sqrt{\mu_{A-BC}} R \quad (30)$$

$$Q_2 = \sqrt{\mu_{BC}} r \quad (31)$$

- Skew angle (β)

$$\beta = \cos^{-1} \left(\sqrt{\frac{m_A m_C}{(m_B + m_C)(m_A + m_B)}} \right) . \quad (32)$$

Q_1 and Q_2 are connected with R and r , respectively.

Chapter 3: Extras

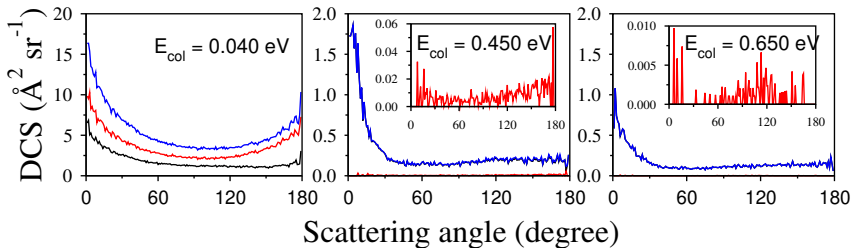


Figure 24: Direct and indirect DCSs for R1 and R2.

Chapter 4: Extras

Comparing fitted and ab initio data

Global minima:

Parameters of global minima	This work	ab initio values
$R_{\text{LiHe}^+} (a_0)$	3.636	3.650
$R_{\text{LiH}^+} (a_0)$	4.192	4.191
$R_{\text{HeH}} (a_0)$	7.829	7.841
$\angle \text{HeLiH}^+$	180°	180°
E (eV)	-0.2038	-0.2034

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E (eV)	-0.2038	-0.2034

Diatomic minima:

	$R_{eq} (a_0)$	$E_{eq} \text{ (eV)}$
LiHe⁺		
This work	3.638	-7.509E-2
Wernli et al.	3.586	-7.965E-2
ab initio data	3.641	
LiH⁺		
This work	4.183	-0.1325
Wernli et al.	4.142	-0.1380
ab initio data	4.186	

Chapter 5: Extras

Different PESs for O₃

2001	R. Siebert, R. Schinke, and M. Bittererova. Spectroscopy of Ozone at the Dissociation Threshold: Quantum Calculations of Bound and Resonance States on a New Global Potential Energy Surface. <i>Phys. Chem. Chem. Phys.</i> 3, 1795 (2001).
2003	P. Fleurat-Lessard, S. Y. Grebenshchikov, R. Siebert, R. Schinke, and N. Halberstadt. Theoretical Investigation of the Temperature Dependence of the O + O ₂ Exchange Reaction. <i>J. Chem. Phys.</i> 118, 610 (2003).
2003	D. Babikov, B. K. Kendrick, R. B. Walker, R. T. Pack, P. Fleurat-Lesard, and R. Schinke. Metastable States of Ozone Calculated on an Accurate Potential Energy Surface. <i>J. Chem. Phys.</i> 118, 6298 (2003).
2004	R. Schinke and P. Fleurat-Lessard. The transition-state region of the O + O ₂ potential energy surface. <i>J. Chem. Phys.</i> 121, 5789 (2004).
2010	Hernández-Lamonedá, Michael R. Salazar, and R.T. Pack. Does ozone have a barrier to dissociation and recombination? <i>Chem. Phys. Lett.</i> 355, 478 (2002).
2010	Filip Holka, Péter G. Szalay, Thomas Müller, and Vladimir G. Tyuterev. Toward an Improved Ground State Potential Energy Surface of Ozone. <i>J. Phys. Chem. A</i> 114, 9927 (2010).
2011	Dawes, R.; Lolur, P.; Ma, J.; Guo, H. Highly Accurate Ozone Formation Potential and Implications for Kinetics. <i>J. Chem. Phys.</i> 135, 081102 (2011).
2013	M. Ayouz and D. Babikov. Global permutationally invariant potential energy surface for ozone forming reaction. <i>J. Chem. Phys.</i> 138, 164311 (2013).
2013	V. G. Tyuterev, R. V. Kochanov, S. A. Tashkun, F. Holka, and P. Szalay. New Analytical Model for the Ozone Electronic Ground State Potential Surface and Accurate ab initio Vibrational Predictions at High Energy Range. <i>J. Chem. Phys.</i> 139, 134307 (2013).
2013	R. Dawes, P. Lolur, A. Li, B. Jiang, and H. Guo. An Accurate Global Potential Energy Surface for the Ground Electronic State of Ozone. <i>J. Chem. Phys.</i> 139, 201103 (2013).
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Chapter 5: Extras

8+66 ICS from 0.01 - 0.1 eV

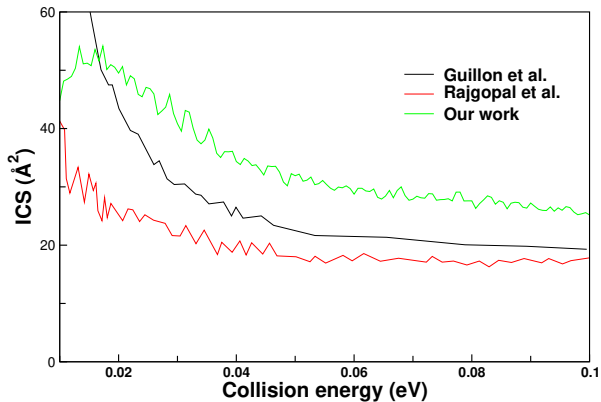


Figure 25: ICS for 8+66 reaction.